

COURSE: CS 352 — Deep Learning

Credits: 3 | Department: Computer Science

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PREREQUISITES: CS 351

CO-REQUISITES: None

OFFERED: Spring

DESCRIPTION:

Neural network architectures and training. Topics include feedforward networks, backpropagation, convolutional neural networks (CNNs), recurrent neural networks (RNNs, LSTMs), attention mechanisms and transformers, generative models (VAEs, GANs, diffusion models), transfer learning, and practical training considerations (optimization, batch normalization, dropout). Projects use PyTorch.

GRADING: Programming projects 50%, Paper review presentations 20%, Final project 30%.